BIO 280 PATHOPHYSIOLOGY · JOYCE UNIVERSITY

Dr. Garg's Exam 3 Master Study Guide

Modules 5 & 6 — Mental Health, Neuro, MSK, Skin & Burns

ⓘ What Exam 3 covers

This unit is about **the brain, the body's framework, and its outer covering**. **Module 5** covers mental health disorders and neurological emergencies — from depression and schizophrenia to strokes, increased intracranial pressure, and spinal cord injuries. **Module 6** covers the musculoskeletal system, skin disorders, and burns. Exam format: **50 questions (multiple choice + select-all-that-apply)**, proctored with Respondus LockDown Browser.

ⓘ How this guide works

- **Click any topic card** in the navigator to jump to that section. Click a title bar to expand or close it.
- **All answers are hidden** — try first, then click "Show answer." Practice includes **multiple choice** and **select-all-that-apply (SATA)**, like the real exam.
- **Each topic has:** clear explanation, a mnemonic, a "common confusion" warning, practice questions, a clinical scenario, and a self-check.
- **Color code: indigo = Module 5 (Mental Health & Neuro), teal = Module 6 (MSK, Skin & Burns)**.
- **To save as PDF:** use the button above, or press *Ctrl+P* / *Cmd+P* → "Save as PDF." All sections expand for printing.

□Suggested 7-Day Study Schedule ▼

Day	Focus
Day 1	Topics 1–2 (mood disorders; anxiety, schizophrenia, ADHD).
Day 2	Topics 3–4 (ICP/Cushing's/herniation/headaches; stroke/TIA/Bell's palsy).
Day 3	Topics 5–6 (TBI/hematomas/hydrocephalus/meningitis; SCI/shock/autonomic dysreflexia).
Day 4	Topics 7–8 (trauma assessment & spinal curves; bone disorders).
Day 5	Topics 9–10 (fracture complications & soft tissue; skin infections).
Day 6	Topics 11–12 (chronic skin & cancers; ulcers & burns).
Day 7	Cheat sheet + comparison tables. Self-test with hidden answers.

□Topic Navigator

TOPIC 1 · MODULE 5
Mood Disorders: Depression & Bipolar

TOPIC 2 · MODULE 5
Anxiety, Schizophrenia & ADHD

TOPIC 3 · MODULE 5
Increased ICP, Cushing's Triad & Headaches

TOPIC 4 · MODULE 5
Stroke, TIA & Bell's Palsy

TOPIC 5 · MODULE 5
TBI, Brain Bleeds, Hydrocephalus & Meningitis

TOPIC 6 · MODULE 5
Spinal Cord Injury, Shock & Seizures

TOPIC 7 · MODULE 6

TOPIC 8 · MODULE 6

TOPIC 9 · MODULE 6

Trauma Assessment & Spinal Curvatures

Bone Disorders

Fracture Complications & Soft Tissue Injuries

TOPIC 10 · MODULE 6

Skin Infections

TOPIC 11 · MODULE 6

Chronic Skin Conditions & Skin Cancers

TOPIC 12 · MODULE 6

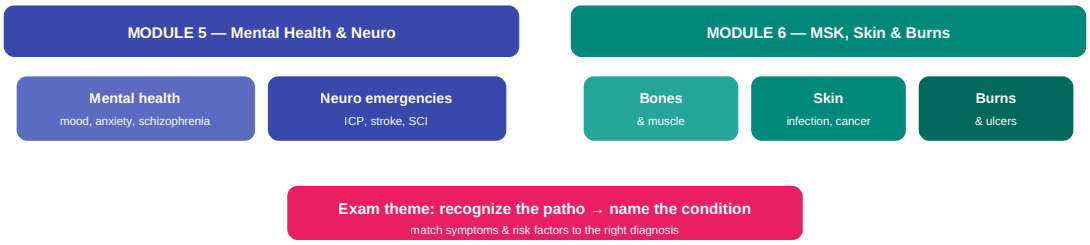
Ulcers & Burns

□Exam Cheat Sheet

□Don't-Mix-These-Up

□Glossary

□The Big Picture: Two Modules



TOPIC 1

☹ Mood Disorders: Depression & Bipolar

MUST KNOW ▼

Depression (Major Depressive Disorder, MDD)

Persistent feelings of **sadness or apathy lasting 2+ weeks**. Common symptoms: low energy, sleep problems, abnormal eating, despair, guilt, hopelessness, and **suicidal ideation**. Subtypes:

- **Postpartum depression** — up to 80% of mothers get mild "baby blues" in the first 2 weeks; ~10–15% develop more severe PPD; linked to hormonal changes.
- **Seasonal Affective Disorder (SAD)** — occurs mostly in **winter** (less sunlight); associated with **decreased melatonin and serotonin**.
- **Dysthymia** — mild depression lasting longer than 2 years.

Causes of Depression (multifactorial)

Category	Examples
Biological	Neurotransmitter imbalance (**serotonin, dopamine, norepinephrine**), hormones, low sunlight/vitamin D.
Genetic	Family history of mood disorders.
Psychosocial	Trauma, grief, chronic stress, isolation.
Medical	Chronic illness, medication side effects, postpartum changes.

> **High-yield safety point:** suicide risk is *highest* when treatment **begins to work** — energy returns before mood fully lifts, so a person now has the energy to act on suicidal thoughts.

Bipolar Disorder

A serious disorder with a **higher-than-average suicide risk**, defined by extreme mood swings between **mania** and **depression**:

- **Mania** — grandiose ideas, **decreased need for sleep/food**, distractibility, impaired judgment (spending, risky sexual behavior), **pressured speech**, possibly delusions/paranoia/hallucinations.
- **Depression** — the classic depressive symptoms above.

- **Cyclothymia** — milder emotional highs and lows.

Mnemonic: Depression neurotransmitters = "**Sad people lack SND**" (Serotonin, Norepinephrine, Dopamine). **SAD = Sun-Absence Depression** (winter, low melatonin/serotonin). Mania = "**DIG FAST**" (Distractibility, Insomnia/↓sleep, Grandiosity, Flight of ideas, Activity↑, Speech pressured, Thoughtlessness/risk).

⚠ **Common confusion:** Suicide risk peaks as a depressed patient **starts to improve** (regains energy before mood lifts) — not at the lowest point. And **bipolar** requires episodes of **mania** (not just depression); without mania it's major depression.

Practice Questions

1. How long must low mood persist to meet criteria for major depressive disorder?

   Show answer

   At least 2 weeks of persistent sadness or apathy.

2. Why does seasonal affective disorder occur mostly in winter?

   Show answer

   Reduced sunlight alters melatonin and serotonin, lowering mood during the dark winter months.

3. **SATA:** Which are signs of a manic episode? (A) Decreased need for sleep (B) Pressured speech (C) Grandiose ideas (D) Excessive guilt and apathy (E) Impulsive spending

   Show answer

   A, B, C, E. Excessive guilt and apathy are depressive, not manic.

4. When is a depressed client at highest risk for suicide?

   Show answer

   When treatment starts to work and energy returns before mood fully improves.

Clinical Scenario

A client with severe depression starts an antidepressant. Two weeks later they seem more energetic and are being closely monitored for safety. Why the extra caution now?

Show answer

Antidepressants often restore **energy and motivation before they lift mood**. A still-hopeless patient who now has energy may be at the **highest risk for acting on suicidal thoughts**, so close monitoring is critical in the early treatment window.

☑ **Self-check:** Can you define MDD (2 weeks), explain SAD, list manic signs (DIG FAST), and state when suicide risk peaks? If yes, move on.

TOPIC 2

Anxiety, Schizophrenia & ADHD

MUST KNOW ▼

Anxiety Disorders

Excessive fear or worry that interferes with daily function — with both psychological AND physiological effects (rapid heart rate, hypertension, restlessness). Types:

Type	Key feature
GAD	Generalized, **persistent** worry and fear.
Social anxiety	Fear of public/social interaction.
OCD	Unwanted, recurrent **behaviors performed to reduce anxiety**.
Panic disorder	Sudden intense emotional unrest triggered by stress, fear, or exertion.
PTSD	Follows a **traumatic event**; varied psychological/physical effects.

Schizophrenia

A chronic, severe disorder that **impairs perception of reality**, thought patterns, and functioning. Linked to **dopamine dysregulation**. Symptoms split into two groups:

Positive (added) — *excess* dopamine	Negative (lost) — *low* dopamine
Hallucinations, delusions, disorganized speech, paranoia (psychosis)	**Alogia** (no speech), **avolition** (no motivation), **anhedonia** (no pleasure), social withdrawal

Positive = adds something abnormal (hallucinations); **negative = takes normal function away** (no motivation).

ADHD

A **neurodevelopmental** disorder usually beginning in childhood (can persist into adulthood), from dysregulation of **dopamine and norepinephrine** (these are *low*). Core symptoms: **inattention, hyperactivity, impulsiveness**. Risk factors: genetics, traumatic brain injury, in-utero toxins (lead, alcohol, cigarette smoke). Associated: low self-esteem, social/academic difficulty.

> **Mnemonic:** Schizophrenia: "**Positive = Presence** (hallucinations/delusions, too much dopamine); **Negative = Nothing** (the 3 A's: Alogia, Avolition,

Anhedonia, too little dopamine)." ADHD = "**low Dopamine & Norepinephrine**" (which is why stimulants that raise them help).

⚠ **Common confusion:** In schizophrenia, **positive symptoms come from EXCESS dopamine** (treated with dopamine blockers) while **negative symptoms come from LOW dopamine**. "Positive" doesn't mean good — it means symptoms *added* to normal experience.

Practice Questions

1. Which anxiety disorder involves recurrent behaviors performed to relieve anxiety?

   Show answer

   Obsessive-compulsive disorder (OCD).

2. **SATA:** Which are POSITIVE symptoms of schizophrenia? (A) Avolition (B) Hallucinations (C) Paranoia (D) Social withdrawal (E) Delusions

   Show answer

   B, C, E. Avolition and social withdrawal are negative symptoms.

3. Which neurotransmitters are dysregulated (low) in ADHD?

   Show answer

   Dopamine and norepinephrine.

4. What are the three core symptoms of ADHD?

   Show answer

   Inattention, hyperactivity, and impulsiveness.

Clinical Scenario

A child with ADHD is prescribed a stimulant medication, which seems counterintuitive ("why give a stimulant to a hyperactive child?"). Using the neurotransmitter basis, explain why it works.

Show answer

ADHD involves **low dopamine and norepinephrine** in brain regions controlling attention and impulse. Stimulants **raise** these neurotransmitters, improving focus and self-control — so they calm symptoms rather than worsen hyperactivity.

☑ **Self-check:** Can you match the anxiety subtypes, split schizophrenia into positive (excess dopamine) vs negative (low dopamine), and give ADHD's neurotransmitters + symptoms? If yes, move on.

TOPIC 3

□Increased ICP, Cushing's Triad & Headaches

MUST KNOW

Increased Intracranial Pressure (ICP)

ICP is pressure inside the rigid skull. The skull holds **3 things** (the Monro-Kellie idea): **brain tissue, blood, and cerebrospinal fluid (CSF)**. Since the skull can't expand, if any one increases, pressure rises and "suffocates" the brain. Causes:

Component ↑	Examples
Blood	Vasodilation (high CO&sub2;), venous obstruction, very high BP, hemorrhage.
Brain tissue	Tumor, edema (from injury/bleed).
CSF	Hydrocephalus (blocked drainage or overproduction).

Early vs. Late Signs

Early	Late (ominous)
Headache, nausea, vomiting, drowsiness, confusion, double vision	**Cushing's Triad**, **fixed/dilated pupils** (CN III compression), decreased LOC, abnormal posturing, coma

Cushing's Triad (late, life-threatening)

Cushing's Triad = ↑ BP (widening pulse pressure) · ↓ HR (bradycardia) · ↓ RR (irregular)

A sign the brainstem is being compressed — a neurological emergency.

Posturing reflects severe damage: **Decorticate = arms flexed** toward the *core* ("de-COR-ticate, arms to the core"); **Decerebrate = arms extended** (worse).

Headache Types

Type	Hallmark

Tension (most common)	**Bilateral** "tight band" around the head; from tight muscles, poor posture, stress.
Sinus	Frontal/below the eyes; **facial pain**, often with a cold/congestion.
Cluster	**Unilateral, stabbing around one eye**; eye tearing, nasal congestion (a type of migraine).
Migraine	Severe, **unilateral, pulsating/throbbing**; nausea/vomiting, photophobia & phonophobia; worse with movement.

Mnemonic: Skull's 3 contents: "**Blood, Brain, CSF.**" Cushing's Triad = "**high BP, low & slow (HR & RR)**" — the opposite of shock. **Decorticate = to the CORE** (flexed); decerebrate = extended (worse). Cluster = "**Clusters around one eye**"; tension = "**Tight band, both sides.**"

⚠ **Common confusion:** **Cushing's Triad** (a sign of high ICP) is high BP + LOW HR + LOW/irregular RR — do NOT confuse it with shock (low BP + high HR). Cushing's Triad is a **late** finding; by the time it appears, herniation may be imminent.

Practice Questions

1. What are the 3 components inside the skull?

   Show answer

   > Brain tissue, blood, and cerebrospinal fluid (CSF).

2. Name the 3 components of Cushing's Triad.

   Show answer

   > Increased BP (widening pulse pressure), decreased heart rate (bradycardia), decreased/irregular respiratory rate.

3. **SATA:** Which are LATE signs of increased ICP? (A) Hypertension (B) Bradycardia (C) Mild headache (D) Decreased respiratory rate (E) Fixed/dilated pupils

   Show answer

   > **A, B, D, E.** Mild headache is an early sign.

4. A patient has stabbing pain around one eye with tearing and nasal congestion. Headache type?

Show answer

Cluster headache.

Clinical Scenario

A head-injury patient who was talking now has a BP of 190/70, HR of 48, and slow irregular breathing. Why is this an emergency, and what's happening?

Show answer

These vitals are **Cushing's Triad** (high BP/widening pulse pressure, bradycardia, slow irregular breathing) — a **late sign of dangerously high ICP with brainstem compression**. The brain is being squeezed and herniation may be imminent; this requires immediate intervention to lower ICP.

☑ **Self-check:** Can you list the skull's 3 contents, recite Cushing's Triad, distinguish early vs late ICP signs, and match the 4 headache types? If yes, move on.

TOPIC 4

□Stroke (CVA), TIA & Bell's Palsy

MUST KNOW ▼

A **stroke (cerebrovascular accident, CVA)** is the **death of brain cells** from interrupted blood flow. There are two fundamentally different types — and telling them apart matters because the treatments are opposite.

Ischemic vs. Hemorrhagic

	Ischemic (80–85%)	Hemorrhagic (15–20%)
Cause	**Blockage** of a vessel — thrombus (from atherosclerosis) or embolus (from A-fib)	**Rupture** of a weakened vessel
Risk factors	Atherosclerosis, atrial fibrillation, clots	**Hypertension (#1)**, aneurysm, trauma
Classic signs	Facial drooping, one-sided weakness, speech changes, confusion	**Sudden severe ("thunderclap") headache**, sudden vomiting or LOC
Treatment	**Fibrinolytics** (clot busters) — only after CT rules out a bleed!	Stop the bleeding / control BP (NO clot busters)

> **Why CT first:** a clot-busting drug given to a hemorrhagic stroke would be catastrophic — so a head CT must rule out a bleed before fibrinolytics.

TIA (Transient Ischemic Attack)

A **temporary** ischemic event where **symptoms fully resolve** with **no permanent damage**. It's a **"warning sign"** of a high likelihood of a bigger stroke. (Modern definition is tissue-based, not strictly time-based.)

Bell's Palsy

Unilateral paralysis of the facial nerve (cranial nerve VII). Cause usually unknown (herpes infection or Lyme suspected). **Facial weakness ONLY — no symptoms in the arms or legs.** This is the key way to tell it from a stroke:

Bell's palsy vs stroke: Bell's palsy affects the **whole side of the face (including the forehead)** with NO limb involvement. A stroke usually **spares the forehead** and often comes with **arm/leg weakness and speech changes**.

Mnemonic: Stroke screen = "**BE FAST**" (Balance, Eyes, Face droop, Arm weakness, Speech, Time). **Ischemic = "I-clog" (clot)**; **Hemorrhagic = "Hemorrhage = bleed," thunderclap headache**, #1 cause HTN. **TIA = "Temporary, a warning."** Bell's palsy = "**face only, no limbs.**"

⚠ **Common confusion:** Don't give **fibrinolytics** until a CT rules out a **hemorrhagic** stroke. And **Bell's palsy involves the forehead and ONLY the face**; if there's limb weakness or forehead sparing, think stroke instead.

Practice Questions

1. What causes an ischemic vs a hemorrhagic stroke?

   Show answer

   Ischemic = a blockage (thrombus/embolus); hemorrhagic = rupture of a weakened vessel (often from hypertension).

2. Why must a CT be done before giving clot-busting drugs?

   Show answer

   To rule out a hemorrhagic stroke — fibrinolytics would worsen a bleed catastrophically.

3. **SATA:** Which are true of a TIA? (A) Symptoms resolve (B) No permanent damage (C) It's a warning of higher stroke risk (D) It is hemorrhagic (E) Requires immediate surgery

   Show answer

   A, B, C. TIA is ischemic (not hemorrhagic) and doesn't require surgery.

4. How can you distinguish Bell's palsy from a stroke?

Show answer

Bell's palsy causes facial weakness only (including the forehead) with no limb involvement; stroke usually spares the forehead and includes arm/leg weakness and speech changes.

Clinical Scenario

A patient arrives with sudden one-sided weakness and slurred speech that started an hour ago. The team rushes a CT before any medication. Explain the reasoning and what treatment follows if the CT is clear.

Show answer

The symptoms suggest a stroke. A **CT rules out a hemorrhagic stroke** first, because giving **fibrinolytics** (clot busters) to a bleed would be deadly. If the CT shows no bleed (consistent with an **ischemic** stroke) and the patient is within the time window, fibrinolytics can be given to dissolve the clot — "time is brain."

☑ **Self-check:** Can you contrast ischemic vs hemorrhagic (cause, signs, treatment), define TIA, and distinguish Bell's palsy from stroke? If yes, move on.

TOPIC 5

TBI, Brain Bleeds, Hydrocephalus & Meningitis

MUST KNOW

Traumatic Brain Injury (TBI)

- **Concussion (mild TBI)** — from impact/whiplash; nausea, vomiting, confusion, blurred vision, impaired coordination, headache, personality changes.
- **Contusion** — a brain "bruise"; **coup-contrecoup** (injury at the impact site AND the opposite side).
- **Diffuse axonal injury** — violent rotation shears axons; often causes unconsciousness or a persistent vegetative state.
- **Penetrating injury** — an object enters the skull.

Brain Bleeds (Hematomas/Hemorrhage) — named by location

Type	Where & speed
Epidural hematoma	**Above the dura** (between skull & dura); **arterial** — bleeds **fast and severely**.
Subdural hematoma	**Below the dura** (between dura & arachnoid); **venous** — usually **slower**.
Intracerebral hemorrhage	**Within the brain tissue** (a type of hemorrhagic stroke); from HTN/ aneurysm/trauma.
Subarachnoid hemorrhage	In the **subarachnoid space**; often a ruptured **berry aneurysm**.

Decode the names: "epi-" = above, "sub-" = below; "dural" refers to the dura mater. **Epidural = Arterial = fast** ("E and A early"); **Subdural = venous = slow.**

Hydrocephalus ("Water on the Brain")

Accumulation of CSF in the ventricles (from blocked drainage — most common — or overproduction). Signs differ by age:

Infant (soft skull)	Adult (fixed skull)
Increased head circumference, LOC changes, irritability	Headache, vomiting, ataxia, urinary incontinence, lethargy, coma

An infant's skull bones aren't fused, so the head enlarges; an adult's fused skull instead develops rising ICP signs.

Meningitis vs. Encephalitis

	Meningitis	Encephalitis
What's inflamed	The **meninges** (membranes)	The **brain tissue** itself
Cause	Bacterial (more severe) or viral (often self-limiting)	Usually viral (**herpes** most common)
Hallmark	**Classic triad: nuchal rigidity (stiff neck), severe headache, fever** + photophobia	Nonspecific: fever, headache, photophobia, **confusion**

Mnemonic: "**Epidural = Artery = fast & deadly; Subdural = vein = slow.**" Meningitis triad = "**Stiff neck, Headache, Fever.**" Hydrocephalus: **infant head grows; adult head can't, so pressure climbs.** Encephalitis = "**encephalo = brain** tissue inflamed (confusion)."

⚠ **Common confusion:** **Meningitis = membranes; encephalitis = brain tissue** (confusion is more prominent in encephalitis). And **epidural** bleeds are **arterial** (rapid deterioration), while **subdural** bleeds are **venous** (slower, common in elderly/alcoholics after falls).

Practice Questions

1. Which brain bleed is arterial and fast, and where is it?

   Show answer

   > Epidural hematoma — arterial bleed above the dura (between skull and dura mater).

2. Why does hydrocephalus increase head circumference in infants but not adults?

   Show answer

   > Infant skull bones aren't fused, so the head expands; an adult's fused skull can't, so ICP rises (headache, vomiting, ataxia) instead.

3. What is the classic triad of meningitis?

Show answer

Nuchal rigidity (stiff neck), severe headache, and fever.

4. A boxer is hit and the brain is injured both at the blow site and the opposite side. What is this called?

Show answer

Coup-contrecoup (a contusion pattern).

Clinical Scenario

After a head injury, a patient is briefly knocked out, wakes up and feels fine ("lucid interval"), then rapidly deteriorates into unconsciousness. Which bleed does this classically suggest, and why the rapid decline?

Show answer

An **epidural hematoma**. It's an **arterial** bleed, so after a brief "lucid interval" the rapidly expanding blood raises ICP quickly and compresses the brain — causing fast deterioration. It's a neurosurgical emergency.

☑ **Self-check:** Can you classify TBIs, locate the 4 brain bleeds (epidural=arterial/fast), explain hydrocephalus by age, and contrast meningitis vs encephalitis? If yes, move on.

TOPIC 6

Spinal Cord Injury, Shock & Seizures

MUST KNOW

Spinal Cord Injury (SCI)

Damage that interrupts **motor, sensory, or autonomic pathways**. Most-injured areas: **cervical and lumbar** (most mobile). The **level** determines the deficit:

- **Quadriplegia** — injury at **C8 and above** (all four limbs). **"C5 keeps the diaphragm alive"** — injury above C5 can stop breathing.
- **Paraplegia** — injury at **T1 and below** (legs). Injuries at **T6 and above** can cause neurogenic shock (bradycardia).

Injury types (by mechanism): **flexion** (head-on MVA), **hyperextension** (hitting chin in a fall), **compression** (diving into shallow water, object on head), **subluxation** (falling off a horse).

Spinal Shock vs. Neurogenic Shock

	Spinal Shock	Neurogenic Shock
What it is	Temporary **loss of reflexes & flaccidity** below injury	Loss of **sympathetic outflow** → hemodynamic instability (injury usually above T6)
Signs	Absent deep tendon reflexes, flaccid paralysis, loss of sensation below injury	**Hypotension + bradycardia** (unopposed parasympathetic tone)
Course	Won't know extent of injury for 4–6 weeks	Often resolves in 24–48 hrs once stabilized

The tell: neurogenic shock's **bradycardia** (slow HR) is unusual — most other shocks cause a fast HR. It's from losing sympathetic ("fight or flight") tone, leaving the vagus nerve unopposed.

Autonomic Dysreflexia (a later SCI complication)

A dangerous **sympathetic overdrive** in patients with SCI at **T6 or higher**, triggered by an **irritant below the injury** (full bladder, constipation, tight clothing). The body can't coordinate its response across the injury:

- **Above the injury:** vasodilation → forehead sweating, facial flushing, pounding headache.
- **Below the injury:** vasoconstriction → **severe hypertension**.
- The brain senses the high BP and tries to compensate with **bradycardia**. Emergency — find and remove the trigger!

Seizures & Spina Bifida

- **Seizure** — sudden, disorganized electrical discharge in the brain. **Status epilepticus** = a seizure lasting **>5 minutes** OR **2+ seizures within 5 minutes** (an emergency).
- **Spina bifida** — improper closing of the spinal column in development, caused by **maternal folate deficiency**; **prevented by taking folate**. Types: occulta (hidden, minimal), meningocele, myelomeningocele (most severe, often lower-limb paralysis).

Mnemonic: "**C5 keeps the diaphragm alive.**" Neurogenic/autonomic dysreflexia both cause **bradycardia + a BP problem** (think "the slow-heart shock"). Autonomic dysreflexia = "**find the irritant below** (bladder, bowel, tight clothes)." **Folate prevents spina bifida.** Status epilepticus = "**5 minutes is too long.**"

⚠ **Common confusion:** Both neurogenic shock and autonomic dysreflexia cause **bradycardia**, but neurogenic shock has **low BP** while autonomic dysreflexia has **severe HIGH BP**. **Spinal shock** (loss of reflexes/flaccidity) is different from **neurogenic shock** (cardiovascular instability).

Practice Questions

1. What does "C5 keeps the diaphragm alive" mean?

   Show answer

   > The diaphragm is innervated around C3–C5 (phrenic nerve); injuries above C5 can paralyze the diaphragm and stop breathing.

2. Distinguish spinal shock from neurogenic shock.

Show answer

Spinal shock = temporary loss of reflexes and flaccidity below the injury; neurogenic shock = loss of sympathetic tone causing hypotension and bradycardia.

3. **SATA:** Which are signs of autonomic dysreflexia? (A) Forehead sweating (B) Severe hypertension (C) Bradycardia (D) Hypotension (E) Pounding headache

   Show answer

   A, B, C, E. Autonomic dysreflexia causes severe HYPERtension, not hypotension.

4. Define status epilepticus.

   Show answer

   A seizure lasting more than 5 minutes, or two or more seizures within 5 minutes without recovery.

Clinical Scenario

A patient with a C5 spinal cord injury suddenly develops a pounding headache, flushing and sweating on the face, and a BP of 210/110. What's happening, and what's the first thing to look for?

Show answer

This is **autonomic dysreflexia** (SCI at T6 or above). An irritant below the injury (most commonly a **full bladder**, then constipation or tight clothing) triggers sympathetic overdrive and dangerous hypertension. Immediately find and remove the trigger (check/empty the bladder first) and sit the patient up to lower BP.

☑ **Self-check:** Can you tie SCI level to quadri-/paraplegia, contrast spinal vs neurogenic shock, explain autonomic dysreflexia (high BP + brady), and define status epilepticus? If yes, you've finished Module 5!

TOPIC 7

Trauma Assessment & Spinal Curvatures MUST KNOW

Primary Survey (ABCDE) — for trauma

Used to **immediately identify life-threatening conditions** in trauma, especially in unresponsive clients. Work through it in order:

Letter	Check
A — Airway	Is the airway occluded?
B — Breathing	Respiratory distress? Normal breath sounds?
C — Circulation	Heart rate, pallor, capillary refill.
D — Disability	Neurologic status — can they respond?
E — Exposure/Environment	Expose to find injuries; environmental (e.g., hypothermia).

Priority Assessment (the 5 P's) — for fractures/dislocations

Used to check the **health of tissue around a fracture or dislocation** — has the bone damaged nearby nerves, arteries, or veins?

P	Meaning
Pain	Can they detect pain?
Pulse	Is blood circulating?
Pallor	Pale = poor perfusion.
Paresthesia	Can they feel (sensation)?
Paralysis	Can they move?

Spinal Curvatures

Curve	Description
Kyphosis	Exaggerated **forward rounding of the upper back** ("hunchback"). Common with osteoporosis.

Lordosis	Exaggerated **inward curve of the lower back** ("swayback").
Scoliosis	**Sideways (lateral) curvature** of the spine.

Mnemonic: Trauma = "**ABCDE**" (Airway first, always). Fracture = "**5 P's**" (Pain, Pulse, Pallor, Paresthesia, Paralysis). Curves: "**Kyphosis = K looks hunched forward; Lordosis = Looking up/swayback; Scoliosis = S-shaped sideways.**"

⚠ **Common confusion:** The **primary survey (ABCDE)** is for whole-body trauma/life threats; the **priority assessment (5 P's)** is for limb/tissue health around a fracture. **Kyphosis = forward hunch, lordosis = inward sway, scoliosis = sideways** — don't mix them up.

Practice Questions

1. What do the letters in the trauma primary survey stand for?

   Show answer

   > Airway, Breathing, Circulation, Disability, Exposure/Environment.

2. List the 5 P's of the priority (neurovascular) assessment.

   Show answer

   > Pain, Pulse, Pallor, Paresthesia, Paralysis.

3. Which spinal curvature is a sideways curve?

   Show answer

   > Scoliosis.

Multiple Choice Practice

1. In the primary survey, what is assessed FIRST?
   A) Circulation · B) Disability · C) Airway · D) Exposure

   Show answer

   > **C.** Airway (the "A" in ABCDE).

2. Exaggerated forward rounding of the upper back is:
   A) Lordosis · B) Scoliosis · C) Kyphosis · D) Stenosis

Show answer

C. Kyphosis.

3. The 5 P's assessment is used for:
   A) Mental status · B) Fractures/dislocations · C) Stroke · D) Burns

   Show answer

   B. Assessing tissue around a fracture or dislocation.

Clinical Scenario

A patient with a casted forearm fracture reports increasing numbness and the fingers look pale with a weak pulse. Using the 5 P's, why is this urgent?

Show answer

Pallor, paresthesia (numbness), and a weak pulse are abnormal findings in the 5 P's neurovascular check — they signal that blood flow and nerve function to the limb are compromised (possible compartment syndrome). This is an emergency requiring immediate evaluation to prevent permanent tissue/nerve damage.

☑ **Self-check:** Can you recite ABCDE, the 5 P's, and the 3 spinal curvatures with their directions? If yes, move on.

TOPIC 8 □**Bone Disorders** MUST KNOW ▼

Four bone disorders that are easy to mix up. The key is the **underlying problem** for each.

Disorder	Core problem & key facts
Osteoporosis	**Demineralization** — osteoclast (breakdown) activity > osteoblast (building). Bones become porous/brittle. **Risk factors:** menopause/low estrogen, age, Caucasian/Asian race, smoking, alcohol, steroids, low calcium, sedentary lifestyle, hyperthyroidism. Signs: bone pain (back), **kyphosis, height loss, fracture risk**.
Osteomalacia	**"Soft bone"** — impaired mineralization from **vitamin D deficiency** (called **rickets** in children). Linked to low calcium/phosphate, high PTH, renal disease. Signs: bone pain, deformities (bowed legs, pigeon breast).
Osteoarthritis (OA)	**Non-inflammatory "wear and tear"** — cartilage erodes, bone rubs on bone. Irreversible, progresses slowly. Pain **WORSENS as the day goes on** / with use; usually unilateral. Risk: age, female, obesity, prior injury, repetitive joint use.
Osteomyelitis	**Bacterial infection IN the bone** — high risk with **open fractures** (skin/environment contact). Signs: **constant bone pain not relieved by rest**, fever, chills, night sweats, warmth/tenderness. Not seen on early x-ray; untreated → **sepsis**.

□**Mnemonic:** "**OsteoporOsis = pOrOus bones** (too few minerals)." "**OsteoMALACIA = Malformed/soft (vitamin D)**; in kids = Rickets." "**OsteoMYELitis = microbes (infection), think MY bone is infected.**" **OA = "Old joints, worse with use"** (wear & tear; remember RA was worse in the morning, OA worse by evening).

⚠ **Common confusion:** **Osteoporosis** = too little bone *quantity* (demineralization); **osteomalacia** = poor bone *quality*/softening (vitamin D). **Osteoarthritis pain worsens through the day** (vs rheumatoid arthritis, which

improves through the day). **Osteomyelitis** is an *infection* — constant pain + fever.

Practice Questions

1. What is the underlying cause of osteomalacia?

   Show answer

   > Vitamin D deficiency → impaired bone mineralization ("soft bone"); it's called rickets in children.

2. How does osteoarthritis pain change over the day, and how does that differ from RA?

   Show answer

   > OA pain worsens as the day/use goes on; rheumatoid arthritis stiffness is worst in the morning and improves with activity.

3. **SATA:** Which are risk factors for osteoporosis? (A) Menopause/low estrogen (B) Smoking (C) Prolonged steroid use (D) Regular weight-bearing exercise (E) Low calcium intake

   Show answer

   > **A, B, C, E.** Weight-bearing exercise *protects* against osteoporosis.

4. Which bone disorder is a bacterial infection, and what's a major risk factor?

   Show answer

   > Osteomyelitis; open fractures (which expose bone to skin/environmental bacteria).

Clinical Scenario

A postmenopausal woman has a vertebral compression fracture from a minor fall, with kyphosis and height loss. Which bone disorder is most likely, and why does menopause increase the risk?

Show answer

Osteoporosis. Estrogen helps maintain bone density; after menopause, estrogen drops, so osteoclast (breakdown) activity outpaces osteoblast (building) activity. Bones become porous and brittle, leading to fragility fractures, kyphosis, and height loss from even minor trauma.

☑ **Self-check:** Can you give the core problem for osteoporosis, osteomalacia, OA, and osteomyelitis, and the OA-vs-RA timing difference? If yes, move on.

TOPIC 9

Fracture Complications & Soft Tissue Injuries

MUST KNOW

Three Dangerous Fracture Complications

Complication	Patho & signs
Compartment syndrome	Pressure inside a muscle compartment gets so high (from edema) it compresses nerves/vessels → ischemia. **6 P's:** severe sudden **Pain**, **Paresthesia**, **Pallor**, **Paralysis**, **Pulselessness**, **Poikilothermia** (cool). Emergency!
Fat embolism	Fat globules from **yellow bone marrow** enter the blood and occlude vessels. **Triad: Confusion** (brain), **Petechiae** (chest/upper torso rash), **Shortness of breath** (lung). Common with long-bone fractures.
Rhabdomyolysis	Rapid **muscle breakdown** (overuse, trauma, crush, statins) releases **myoglobin + potassium** into the blood. **Triad: Myalgia** (muscle pain), **brown urine** (myoglobinuria), **weakness**. Labs: high creatine kinase, hyperkalemia. Can cause kidney injury.

Osteomyelitis (Topic 8) is also a fracture complication — the infection risk from **open fractures**.

Soft Tissue Injuries

Injury	What's torn/stretched
Sprain	A **ligament** (bone-to-bone) — from a quick twist/pull; signs: bruising, swelling, instability, painful movement. (Common at joints, e.g., ankle.)
Strain	A **muscle or tendon** — common in large muscle groups (low back, calf, hamstring).
Rotator cuff injury	Shoulder; pain with **abduction**, popping/clicking, muscle weakness.
ACL tear	Knee ligament; classic **"plant and twist"** non-contact injury; increases risk of later osteoarthritis.

Mnemonic: Compartment syndrome = "**6 P's**" (the 5 P's + Poikilothermia). Fat embolism triad = "**Confused, Spotty, Breathless**" (brain, petechiae, lungs).

Rhabdo = "**Muscle pain + brown pee.**" Sprain vs strain: "**Sprain = ligament (s for skeletal joint); Strain = muscle/tendon (t for tendon).**"

⚠ **Common confusion: Sprain = ligament; strain = muscle/tendon.** Compartment syndrome's pain is **severe and out of proportion**, with the 6 P's; don't dismiss it as normal fracture pain. **Brown urine after a crush/overuse injury = rhabdomyolysis** (myoglobin), a clue for hyperkalemia and kidney risk.

Practice Questions

1. What is the classic triad of a fat embolism?

   Show answer

   Confusion, petechiae (upper torso), and shortness of breath.

2. What's the difference between a sprain and a strain?

   Show answer

   A sprain is a stretched/torn ligament; a strain is a stretched/torn muscle or tendon.

3. **SATA:** Which are signs of compartment syndrome? (A) Severe sudden pain (B) Paresthesia (C) Pallor (D) Strong bounding pulse (E) Pulselessness

   Show answer

   A, B, C, E. Compartment syndrome causes *pulselessness*, not a strong pulse.

4. What turns the urine brown in rhabdomyolysis, and what dangerous electrolyte is released?

   Show answer

   Myoglobin (from broken-down muscle) darkens the urine; potassium is released, risking hyperkalemia.

Clinical Scenario

A few days after a femur fracture, a patient suddenly becomes confused and short of breath, with a pinpoint rash across the chest. What complication is this, and why does a long-bone fracture cause it?

Show answer

A **fat embolism**. Long bones (like the femur) contain **yellow (fatty) marrow**; a fracture can release fat globules into the bloodstream, which travel to the lungs (dyspnea), brain (confusion), and skin (petechiae) — the classic triad.

☑ **Self-check:** Can you give the 6 P's, the fat embolism triad, the rhabdo triad, and the sprain-vs-strain distinction? If yes, move on.

TOPIC 10

Skin Infections (Bacterial, Viral & Fungal)

MUST KNOW

Bacterial Skin Infections

Most are caused by **Staphylococcus** or **Streptococcus**. The key distinction is **how deep** they go:

Infection	Depth & hallmark
Impetigo	**Superficial**; classic **honey-colored crusts**; very **contagious** (common in children).
Cellulitis	**Deep** infection of the **dermis & subcutaneous tissue (hypodermis)**; red, warm, swollen, tender; drainage is contagious. Can spread to blood (sepsis).

Depth matters: impetigo = surface (epidermis); cellulitis = deep (dermis + subcutaneous). That's why cellulitis is more dangerous.

Viral Skin Infections

- **Herpes Simplex Virus (HSV)** — painful **vesicular (blister) rash**, often with tingling/numbness before a flare. **HSV-1** = oral ("cold sores"); **HSV-2** = genital/sexual contact. Can spread even when asymptomatic; can pass to a baby during delivery (often a C-section is done).
- **Measles** — a **respiratory virus** that affects the skin: a **maculopapular rash that starts on the head and descends**. Plus high fever (104°F), cough, conjunctivitis, coryza (runny nose), and **Koplik spots** (white spots inside the cheeks — pathognomonic).

Fungal Skin Infections

Superficial fungi (e.g., **tinea** "ringworm," **candida**) love **warm, moist areas** (skin folds, feet, groin). Deep fungal infections occur mainly in **immunocompromised** patients.

Mnemonic: "**Impetigo = honey crust, surface; Cellulitis = deep & dangerous.**" Measles = the **"3 C's + Koplik"** (Cough, Coryza, Conjunctivitis +

Koplik spots), rash **head-to-toe**. **HSV-1 = mouth (1 mouth); HSV-2 = below the belt.** Fungus loves "**warm & moist.**"

⚠ **Common confusion: Impetigo is superficial (honey crusts); cellulitis is deep** (dermis/subcutaneous). **Koplik spots** (inside the mouth) are the giveaway for **measles**. And both impetigo and cellulitis are usually **Staph/Strep** — bacterial, not viral.

Practice Questions

1. Which bacterial skin infection is deep (dermis/subcutaneous) and which is superficial with honey crusts?

   Show answer

   Cellulitis is deep; impetigo is superficial with honey-colored crusts.

2. What are Koplik spots a sign of?

   Show answer

   Measles (white spots inside the mouth/cheeks).

3. Differentiate HSV-1 from HSV-2.

   Show answer

   HSV-1 is typically oral (cold sores); HSV-2 is typically genital (sexual/skin-to-skin contact).

4. Where do fungal skin infections like to grow?

   Show answer

   Warm, moist areas (skin folds, feet, groin).

Clinical Scenario

A patient has a red, hot, swollen, tender lower leg with poorly defined borders, spreading over a day, plus fever. Which infection is this, and why is it more concerning than impetigo?

Show answer

Cellulitis. Unlike superficial impetigo, cellulitis is a **deep** infection of the dermis and subcutaneous tissue. Because it's deep and can spread into the bloodstream, it risks **sepsis** and needs prompt antibiotics — the fever signals possible systemic spread.

☑ **Self-check:** Can you separate impetigo (superficial) vs cellulitis (deep), recall measles' signs (3 C's + Koplik), distinguish HSV-1/2, and say where fungi grow? If yes, move on.

TOPIC 11

Chronic Skin Conditions & Skin Cancers MUST KNOW

Chronic / Inflammatory Skin Conditions

Condition	Patho & hallmark
Eczema (atopic dermatitis)	Chronic inflammatory; **dry, itchy, red** skin. Common in childhood; classically on **flexor surfaces** (behind the knees/popliteal, inner elbow/ antecubital). Linked to **asthma & allergies**.
Urticaria (hives)	Itchy **"wheal and flare"** rash; **IgE-mediated histamine release** — seen with **allergic reactions**.
Acne vulgaris	Inflammatory disorder of the **pilosebaceous unit**: increased **sebum**, follicle blockage by keratin, and bacteria (**Propionibacterium / Cutibacterium**).
Psoriasis	Chronic autoimmune; thick, silvery, scaly plaques (rapid skin-cell turnover).

Skin Cancers

Cancer	Behavior
Basal cell carcinoma	Most common; **rarely metastasizes** (least dangerous).
Squamous cell carcinoma	Can **metastasize** to nodes/organs.
Melanoma	**Most dangerous**; high risk of metastasis. Watch the **ABCDEs**.

Melanoma: the ABCDEs

Letter	Warning sign
A — Asymmetry	One half doesn't match the other.
B — Border	Irregular, ragged, or blurred edges.
C — Color	Varied colors within the lesion.
D — Diameter	Larger than ~6 mm (pencil eraser).
E — Evolving	Changing in size, shape, or color over time.

Mnemonic: Melanoma = "**ABCDE**" (Asymmetry, Border, Color, Diameter, Evolving). Cancer danger order: "**Basal = best (rarely spreads), Melanoma = most malignant.**" **Eczema = flexor surfaces + itch + asthma/allergy**; **Urticaria = IgE/histamine hives** (allergy). Acne = "**sebum + plug + bacteria.**"

⚠ **Common confusion: Basal cell = most common but least likely to spread; melanoma = less common but most deadly. Urticaria is IgE/histamine-mediated** (think allergy), while eczema is a chronic inflammatory condition on **flexor surfaces**.

Practice Questions

1. What do the ABCDEs of melanoma stand for?

   Show answer

   Asymmetry, Border irregularity, Color variation, Diameter (>6 mm), Evolving.

2. Which skin cancer is most dangerous, and which is most common but rarely spreads?

   Show answer

   Melanoma is most dangerous; basal cell carcinoma is most common but rarely metastasizes.

3. What mediates urticaria, and when is it commonly seen?

   Show answer

   It's IgE-mediated histamine release ("wheal and flare"), commonly with allergic reactions.

4. Where does eczema classically appear, and what conditions is it linked to?

   Show answer

   On flexor surfaces (behind the knees, inner elbows); linked to asthma and allergies.

Clinical Scenario

A patient points out a mole that's now asymmetric, has ragged borders, several colors, is 8 mm wide, and has changed over a few months. What should the nurse suspect, and why the urgency?

Show answer

The mole hits every **ABCDE** warning sign — highly suspicious for **melanoma**. Melanoma has a high risk of metastasis, so early recognition and biopsy/ removal are critical; delay allows it to spread and become far harder to treat.

☑ **Self-check:** Can you recall the ABCDEs, rank the 3 skin cancers by danger, and describe eczema, urticaria, and acne? If yes, move on.

TOPIC 12 □Ulcers & Burns MUST KNOW ▼

Pressure Ulcers vs. Venous Stasis Ulcers

	Pressure Ulcer (decubitus)	Venous Stasis Ulcer
Cause	**Ischemia from unrelieved pressure** over a bony area → skin breakdown.	**Dysfunctional vein valves** → fluid & waste accumulate in the legs → edema and tissue damage.
Risk factors	Immobility, pressure, shearing, friction, moisture; poor nutrition, aging, steroids.	Chronic venous insufficiency, leg swelling, varicose veins.
Typical location	Over **bony prominences** (sacrum, heels, hips).	Lower legs (around the ankles).

Pressure ulcer factors split into **extrinsic** (pressure, shear, friction, moisture) and **intrinsic** (immobility, poor nutrition, chronic illness, aging).

Burns — Depth Classification

Degree	Depth & appearance
Superficial (1st degree)	**Epidermis only**; red, painful, no blisters (like a sunburn).
Partial thickness (2nd degree)	Epidermis + part of **dermis**; red, **blistered**, very painful.
Full thickness (3rd degree)	Through the **entire dermis** (into fat); white/charred/leathery; **often painless** in the center (nerves destroyed).

Burn Complications

- **Infection / sepsis** — the skin barrier is gone.
- **Hypovolemic shock** — massive fluid loss/leak → edema, hemodynamic instability, and **pre-renal acute kidney injury**.
- **Hypothermia** — loss of temperature regulation.
- **Respiratory dysfunction** — from inhalation injury.
- **Increased metabolic demand** (hypermetabolic state).

Mnemonic: "**Pressure ulcer = Pressure + bony spot; Venous ulcer = Valves fail in the legs.**" Burn depth: "**1st = red, 2nd = blisters, 3rd = no feeling (nerves gone).**" Burn's biggest early killer = **hypovolemic shock** (fluid loss); biggest later killer = **infection/sepsis.**

⚠ **Common confusion:** A **full-thickness (3rd degree) burn is often painLESS** in the center because the nerve endings are destroyed — so lack of pain does NOT mean a minor burn. **Pressure ulcers** are from pressure/ischemia over bone; **venous stasis ulcers** are from failing vein valves in the legs.

Practice Questions

1. What causes a pressure ulcer vs a venous stasis ulcer?

   Show answer

   Pressure ulcer = ischemia from unrelieved pressure over a bony area; venous stasis ulcer = dysfunctional vein valves causing fluid/waste buildup and tissue damage in the legs.

2. Why is a full-thickness (3rd degree) burn often painless centrally?

   Show answer

   The nerve endings in the dermis are destroyed, so the center doesn't transmit pain.

3. **SATA:** Which are complications of major burns? (A) Infection/sepsis (B) Hypovolemic shock (C) Hypothermia (D) Weight gain (E) Acute kidney injury

   Show answer

   A, B, C, E. Burns cause a hypermetabolic state and weight loss, not gain.

4. Which burn depth has blisters and is very painful?

   Show answer

   Partial-thickness (2nd degree) — epidermis + part of dermis.

Clinical Scenario

A patient with extensive burns develops low blood pressure, low urine output, and rising kidney values within hours. Connect the burn to these findings.

Show answer

Major burns cause huge **fluid shifts and plasma loss** through damaged skin → **hypovolemic shock** (low BP). Reduced blood flow to the kidneys causes **pre-renal acute kidney injury** (low urine output, rising creatinine). This is why aggressive fluid resuscitation is a top early priority in burn care.

☑ **Self-check:** Can you contrast pressure vs venous stasis ulcers, classify the 3 burn depths, and list the major burn complications? If yes — all 12 topics done!

Exam Cheat Sheet

MODULE 5 — Mental Health & Neuro

- **Depression:** 2+ weeks; low serotonin/NE/dopamine. **Suicide risk highest as treatment starts to work.** SAD = winter/low melatonin.
- **Bipolar:** mania (DIG FAST) + depression; high suicide risk. **Schizophrenia:** Positive = excess dopamine (hallucinations); Negative = low dopamine (3 A's: alogia, avolition, anhedonia).
- **ADHD:** low dopamine & norepinephrine; inattention, hyperactivity, impulsiveness.
- **Increased ICP:** skull holds brain + blood + CSF. Early = headache/N&V; Late = **Cushing's Triad (↑BP, ↓HR, ↓RR)**, fixed pupils. Decorticate = flexed (to core); decerebrate = extended.
- **Headaches:** tension (bilateral band), sinus (facial), cluster (one eye, stabbing), migraine (unilateral throbbing + photophobia).
- **Stroke:** Ischemic (80%, clot, give fibrinolytics AFTER CT) vs Hemorrhagic (rupture, #1 = HTN, thunderclap HA). **TIA = temporary, a warning.** Bell's palsy = face only (incl. forehead), no limbs.
- **Brain bleeds:** Epidural = arterial/fast (above dura); Subdural = venous/slow (below dura). Meningitis triad = stiff neck + headache + fever; encephalitis = brain tissue (confusion). Hydrocephalus: infant head enlarges, adult ↑ICP.
- **SCI:** "C5 keeps diaphragm alive." Spinal shock = flaccid/no reflexes; Neurogenic shock = ↓BP + ↓HR (above T6). **Autonomic dysreflexia** = severe HTN + bradycardia from irritant below injury (full bladder). Status epilepticus = seizure >5 min. Spina bifida = folate deficiency.

MODULE 6 — MSK, Skin & Burns

- **Trauma:** Primary survey = **ABCDE**. Fracture neurovascular check = **5 P's** (Pain, Pulse, Pallor, Paresthesia, Paralysis). Curves: kyphosis (forward), lordosis (sway), scoliosis (sideways).
- **Bone:** Osteoporosis = demineralization (menopause/steroids/low Ca). Osteomalacia = soft bone (vit D; rickets in kids). OA = wear & tear, worse through the day. Osteomyelitis = bone infection (open fracture → sepsis).
- **Fracture complications:** Compartment syndrome (6 P's). Fat embolism triad = confusion + petechiae + dyspnea. Rhabdomyolysis = myalgia + brown urine + weakness (↑K^+, ↑CK).
- **Soft tissue:** Sprain = ligament; Strain = muscle/tendon. Rotator cuff = pain with abduction. ACL = plant-and-twist.
- **Skin infections:** Impetigo (superficial, honey crust) vs Cellulitis (deep, dermis/subcutaneous). HSV-1 oral, HSV-2 genital. Measles = 3 C's + Koplik spots, rash head-to-toe. Fungi love warm/moist.
- **Chronic skin:** Eczema (flexor surfaces, asthma/allergy), Urticaria (IgE/histamine hives), Acne (sebum + plug + bacteria). Cancers: Basal (common, rarely spreads) < Squamous < Melanoma (deadliest, **ABCDE**).
- **Ulcers:** Pressure (ischemia over bony spot) vs Venous stasis (failing leg vein valves). **Burns:** 1st = red; 2nd = blisters; 3rd = painless/charred. Complications: hypovolemic shock, infection/sepsis, AKI, hypothermia.

□Don't-Mix-These-Up

High-yield distinctions

Pair	Key difference
Cushing's Triad vs shock	Cushing's = ↑BP + ↓HR + ↓RR (high ICP); shock = ↓BP + ↑HR.
Ischemic vs hemorrhagic stroke	Clot (give clot-buster after CT) vs bleed (HTN, thunderclap HA).
Epidural vs subdural	Arterial/fast vs venous/slow.
Neurogenic shock vs autonomic dysreflexia	Both bradycardic, but ↓BP vs severe ↑BP.
Osteoporosis vs osteomalacia	Too little bone (minerals) vs soft bone (vitamin D).
OA vs RA timing	OA worse through the day; RA worse in the morning.
Sprain vs strain	Ligament vs muscle/tendon.
Impetigo vs cellulitis	Superficial honey crust vs deep dermal infection.
Basal cell vs melanoma	Common/rarely spreads vs deadliest/ABCDE.

□Glossary

Mental Health & Neuro Terms

- **Positive vs negative symptoms** — added (psychosis) vs lost (the 3 A's) function in schizophrenia.
- **ICP** — intracranial pressure (brain + blood + CSF in the skull).
- **Cushing's Triad** — ↑BP, ↓HR, ↓RR; late sign of high ICP.
- **Decorticate / Decerebrate** — flexed (to core) / extended posturing.
- **Uncal herniation** — temporal lobe pushed through the tentorial notch, compressing the brainstem.
- **CVA / TIA** — stroke (cell death) / transient ischemic attack (temporary, resolves).
- **Epidural / Subdural** — arterial bleed above dura / venous bleed below dura.
- **Nuchal rigidity** — stiff neck (meningitis).
- **Autonomic dysreflexia** — SCI emergency: severe HTN + bradycardia from an irritant.
- **Status epilepticus** — seizure >5 min or repeated without recovery.

MSK, Skin & Burn Terms

- **Primary survey (ABCDE)** — trauma life-threat assessment.
- **5 P's** — Pain, Pulse, Pallor, Paresthesia, Paralysis (fracture neurovascular check).
- **Kyphosis / Lordosis / Scoliosis** — forward / inward / sideways spinal curves.
- **Osteoporosis / Osteomalacia** — demineralized / soft (vit D) bone.
- **Osteomyelitis** — bone infection.
- **Compartment syndrome** — high compartment pressure (6 P's).
- **Fat embolism** — marrow fat in blood (confusion, petechiae, dyspnea).
- **Rhabdomyolysis** — muscle breakdown (myoglobinuria, ↑K^+, ↑CK).
- **Sprain / Strain** — ligament / muscle-tendon injury.
- **Impetigo / Cellulitis** — superficial honey-crust / deep dermal infection.
- **Urticaria** — IgE/histamine hives. **ABCDE** — melanoma warning signs.
- **Decubitus / Venous stasis ulcer** — pressure ulcer / failing-vein leg ulcer.

□Final Study Tips

- **This exam is pattern recognition** — a scenario describes a patho or symptom set, and you name the condition. Drill the **one hallmark** for each disorder.
- **Memorize the lettered/numbered sets:** Cushing's Triad, ABCDE (trauma) & ABCDE (melanoma), the 5 P's and 6 P's, the fat embolism and rhabdo triads, the meningitis triad.
- **Lock in the "don't mix these up" table** — especially the bradycardia pair (neurogenic shock vs autonomic dysreflexia) and the two bleed types (epidural vs subdural).
- **Watch the vital-sign reversals:** Cushing's Triad and autonomic dysreflexia both look "backwards" (high BP with slow HR).
- **Practice SATA carefully** — read each option as its own true/false statement; partial credit usually isn't given.
- **Use the cheat sheet** the night before AND the morning of the exam, then self-test with the answers hidden. Format: 50 questions, Respondus LockDown Browser — prep your space ahead of time.

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You've got this. Learn each condition's one hallmark, and the scenarios practically answer themselves.

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